Econometrics Midterm

R Functions · 5 code exhibits · W2, W5, W6 · 10 questions

Exhibit A · Lab2.R — descriptive statistics (W2)

Used in questions 1–2

```r
practice <- read.csv("Lab2_Practice.csv", header=TRUE, na.strings=".")
competence <- practice$COMPETENCE
breaks <- practice$BREAKS
grade <- practice$GRADE
hist(competence, breaks=15)
summary(breaks)
sum((competence-mean(competence))^2)/(length(competence)-1)
var(competence)
quantile(breaks)
IQR(breaks)
boxplot(grade)
abline(h=median(grade), col="red")
par(mfrow = c(1, 2))
boxplot(grade)
vioplot(grade)
```

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1. [Exhibit A] Line 9. What does quantile() do?

2. [Exhibit A] Line 13. What does par(mfrow = c(1, 2)) do?

Exhibit B · Lab4.R — functions and strings (W2)

Used in questions 3–4

```
1  library(readxl)
2  library(stringr)
3  to12 <- function(str){
4    hour=as.integer(str)
5    if (hour>12) {
6      return(paste(hour-12, "PM"))
7    } else {
8      return(paste(hour, "AM"))
9    }
10 }
11 hasMultipleLoc <- function(str){
12   str_detect(str, fixed(","))
13 }
14 data=read_excel("Café.xlsx")
15 data <- as.data.frame(data)
16 colnames(data)=c("type","location","openingHour","closingHour")
17 (openingHour <- sapply(data$openingHour, to12))
18 data$openingHour=openingHour
19 (locationText <- ifelse(sapply(data$location, hasMultipleLoc), "Yes", "No"))
20 data$hasMultipleLoc=locationText
```

3. [Exhibit B] Line 12. What does str_detect() do, and what is fixed() for?

4. [Exhibit B] Line 17. What does sapply() do?

Exhibit C · regression.R (condensed) — OLS, inference and prediction (W5)

Used in questions 5–6

```
social <- read.csv("social.csv", header=T)
perform=social$cryp_scoreFINAL
eye=social$RMET_mean
skill=social$Smean_mean
ravens=social$ravens_mean
plot(perform ~ eye)
eye_regr=lm(perform ~ eye)
abline(eye_regr, col="blue")
abline(h=mean(perform), col="red")
abline(v=mean(eye), col="green")
skill_regr=lm(perform ~ skill)
skill_summary=summary(skill_regr)
confint(skill_regr, level=0.99)
(skill_coef=cor(perform,skill)^2)
round(skill_summary$r.squared-skill_coef,16)
mult_lm=lm(perform ~ skill + ravens)
(mult_summary=summary(mult_lm))
predict(mult_lm, data.frame(skill=c(0.5,1.5), ravens=c(1.5,0.5)), interval="confidence",
level=0.95)
points <- seq(min(eye)-3, max(eye)+3, length.out=10000)
pred <- predict(eye_regr, data.frame(eye=points), interval="confidence", level=0.95)
polygon(c(rev(points), points), c(rev(pred[ ,3]), pred[ ,2]), col=rgb(0,0,0,0.1),
border=NA)
```

5. [Exhibit C] Line 13. What does confint() do, and what does level = 0.99 change?

6. [Exhibit C] Line 18. What does predict() do, and what does interval = "confidence" add?

Exhibit D · simple regression demonstration.R (condensed) — simulation of tests and intervals (W5)

Used in questions 7–8

```r
set.seed(4201)
slope=0.2; intercept=3; sampleSize=10; sampleCnt=10000
estimator=c(); t=c(); pValue=c()
for(rowIdx in 1:sampleCnt) {
  fertilizer=rnorm(sampleSize,10,2)
  yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
  coef=summary(lm(yield~fertilizer))$coefficients
  estimator[rowIdx]=coef[2,1]
  t[rowIdx]=coef[2,3]
  pValue[rowIdx]=coef[2,4]
}
hist(estimator)
alpha=0.05
rejection=ifelse(pValue<=alpha, "Reject", "Fail to Reject")
(result=table(rejection))
barplot(result, main="Null Hypothesis Rejection")
sampleSize=12; slope=2; sampleCnt=20
results=data.frame(index=1:sampleCnt)
for(rowIdx in 1:sampleCnt) {
  fertilizer=rnorm(sampleSize,10,2)
  yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
  fer_regr=lm(yield ~ fertilizer)
  results[rowIdx,"slope"]=fer_regr$coefficients[2]
  interval=confint(fer_regr)
  lowerBound=interval["fertilizer","2.5 %"]
  upperBound=interval["fertilizer","97.5 %"]
  results[rowIdx,"inside"]=lowerBound<=slope && slope<=upperBound
}
```

7. [Exhibit D] Line 1. What does set.seed() do?

8. [Exhibit D] Line 14. What does ifelse() do?

Exhibit E · W6 — dummy variables and interactions (lecture examples, Wooldridge Ch.7)

Used in questions 9–10

```r
m1 <- lm(wage ~ female + educ, data = df)
m2 <- lm(wage ~ educ*female, data = df)
m3 <- lm(wage ~ educ + as.factor(region), data = df)
library(car)
linearHypothesis(m2, c("female = 0", "educ:female = 0"))
restricted <- lm(wage ~ educ + exper, data = df)
unrestricted <- lm(wage ~ (educ + exper)*female, data = df)
anova(restricted, unrestricted)
lpm <- lm(inlf ~ educ + age, data = df)
```

9. [Exhibit E] Line 3. What does as.factor() do, and what does lm() then create from it?

10. [Exhibit E] Line 8. What does anova() do when it is given two models?

Econometrics Midterm — Model answers

1. W2 · quantile()
quantile(): returns sample quantiles, the values below which given shares of the data fall
By default it gives five numbers, the 0%, 25%, 50%, 75% and 100% points: the **minimum**, the first **quartile**, the median, the third quartile and the **maximum**
Other points can be asked for with probs, for example quantile(breaks, probs = 0.9)
Key results: quartile · 0% · minimum

2. W2 · par(mfrow =)
par(): sets **graphics parameters** for the plots that follow
mfrow = c(1, 2) splits the plotting window into **1 row and 2 columns**, filled row by row
So the next two plots (lines 14–15) appear **side by side**, until the layout is reset with par(mfrow = c(1, 1))
Key results: graphics parameter · 1 row and 2 columns · side by side

3. W2 · str_detect()
str_detect(): checks whether each string **contains** a pattern and returns **TRUE** or FALSE for each one
fixed(): makes the pattern a **literal** string, not a regular expression
So hasMultipleLoc("Seoul, Suwon") is TRUE and hasMultipleLoc("Seoul") is FALSE
Key results: TRUE · contains · literal

4. W2 · sapply()
sapply(): **applies** a function to **each element** of a vector or list
It then simplifies the results into a **vector** (or a matrix) when it can
Here it runs to12() on every opening hour, which to12() cannot do by itself because its if() takes one value at a time
Key results: each element · applies · vector

5. W5 · confint()
confint(): returns a **confidence interval** for each **coefficient** of a fitted model, a lower and an upper bound per term
By default the level is 95%
level = 0.99 gives 99% intervals, which are **wider**
Key results: confidence interval · coefficient · wider

6. W5 · predict()
predict(): plugs **new data**, a data frame with the model's variable names, into the fitted equation and returns the **predicted** values, here about 0.3475 and 0.3441
interval = "confidence" adds the **lower and upper** bounds, so the result has the columns fit, lwr and upr
Key results: predicted · new data · lower

7. W5 · set.seed()
set.seed(): sets the starting point (seed) of the **random number generator**
Every run with the same seed draws the **same random numbers**, so the simulation is **reproducible**
Without it, rnorm() would give different samples on each run
Key results: reproducible · random number generator · same random numbers

8. W5 · ifelse()
ifelse(): checks a **condition** for **each element** of a vector
Where it is TRUE it returns the **second** argument, and where it is FALSE the third
Here every p-value becomes "Reject" (p-value at most 0.05) or "Fail to Reject"
Key results: condition · each element · second

9. W6 · as.factor()
as.factor(): converts a variable into a **factor**, a **categorical** variable with levels, instead of a number
lm(): then creates one **dummy** for each level except the **base** (reference) level
Each dummy coefficient is the difference from the base level
Key results: categorical · dummy · base

10. W6 · anova()
anova(): **compares** two **nested** models with an **F-test**
It uses the residual sum of squares (**RSS**) of each model and the number of extra parameters, and reports F and its p-value
Here the models differ by female and its interactions, so it is the **Chow** test
Key results: nested · F-test · RSS · Chow